National Technical University of Ukraine
“Igor Sikorsky Kyiv Polytechnic Institute”
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Laboratory work № 6

“Introduction to Artificial Intelligence”

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LABORATORY WORK #6

Neuro-fuzzy modeling

The purpose of the work:obtaining and consolidating knowledge about modeling methods and principles of functioning of neurofuzzy systems, as well as formation of practical skills in the construction of neurofuzzy networks.

Individual tasks

1. To formulate a task in the field of computer technology, for the solution of which the use of a hybrid neural fuzzy network would be justified.
2. Form a sample for training a hybrid neural network.
3. Generate and visualize the structure of a hybrid neural network.
4. To train a hybrid neural network, at the same time to set and justify the parameters of its training.
5. Build a fuzzy inference system for the resulting hybrid neural network.
6. Check the adequacy of the constructed fuzzy model of the hybrid network.
7. Make a report on laboratory work.

Full Code:

In [1]:
```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl
import pandas as pd
from IPython.display import display

# Örnek bir veri seti oluştur
np.random.seed(42)  # Aynı veriyi üretebilmek için seed kullanıyoruz
data = {
    'The first entrance': np.random.randint(0, 1000, size=100),
    'Second entrance': np.random.randint(0, 1000, size=100),
    'The third entrance': np.random.randint(0, 1000, size=100),
    'Day off': np.random.randint(0, 1000, size=100),
}

# Veri setini DataFrame'e dönüştür
df = pd.DataFrame(data)

# Veri setini CSV dosyasına kaydet
df.to_csv('trainDataset.csv', sep=';', index=False)

# Fuzzy mantık işlemleri
entrance1 = ctrl.Antecedent(np.arange(0, 1000, 1), 'entrance1')
entrance2 = ctrl.Antecedent(np.arange(0, 1000, 1), 'entrance2')
entrance3 = ctrl.Antecedent(np.arange(0, 1000, 1), 'entrance3')
day_off = ctrl.Antecedent(np.arange(0, 1000, 1), 'day_off')
output = ctrl.Consequent(np.arange(0, 1000, 1), 'output')
```

```
entrance1.automf(3)
entrance2.automf(3)
entrance3.automf(3)
day_off.automf(3)
output.automf(3)

entrance1['poor'] = fuzz.trimf(entrance1.universe, [0, 0, df['The first entrance
entrance2['poor'] = fuzz.trimf(entrance2.universe, [0, 0, df['Second entrance'].
entrance3['poor'] = fuzz.trimf(entrance3.universe, [0, 0, df['The third entrance
day_off['poor'] = fuzz.trimf(day_off.universe, [0, 0, df['Day off'].values[0]])

rule1 = ctrl.Rule(entrance1['poor'] | entrance2['poor'] | entrance3['poor'] | day

fis = ctrl.ControlSystem([rule1])
fuzzy_system = ctrl.ControlSystemSimulation(fis)

# View the system
display(fis.view())

input("Press Enter to exit...")
```

RESULT:

```
Press Enter to exit...
```

None

```
Press Enter to exit...
''
```